

Unit: HTTP GET vs POST & Servlet Methods

Video Title: #8 Differences between Http Methods GET and POST || Servlet doGet() & doPost() methods

Video Link: <https://youtu.be/Qi1Rdoy0XhI?si=pXp7z7y7uHMIXKXA>

1. Introduction to HTTP Methods

Definition

HTTP Methods: HTTP methods, also called **HTTP verbs**, are used by a client to communicate with a server and tell the server what action should be performed on a resource.

A **client** can be a web browser, mobile application, or another program.

A **server** contains resources such as:

- HTML files
- Images
- Documents
- Java programs
- Other application resources

The client uses a specific **URL** and an HTTP method to communicate with the server.

Common HTTP Methods

Two important HTTP methods used in Servlet programming are:

1. **GET**
 2. **POST**
-

2. GET Method

Definition

GET: GET is an HTTP method mainly used to **request or retrieve information from a server**.

For example, when you search for something on a website, the search information may be sent using a GET request.

How GET Sends Data

In a GET request, data is generally attached to the **URL** as query parameters.

Example:

```
https://example.com/search?name=java
```

Here:

- `search` is the requested resource/path.
- `name=java` is the data sent to the server.
- The data is visible in the URL.

Characteristics of GET

- Data is visible in the **URL**.
- It is generally suitable for **small amounts of data**.
- It is commonly used when requesting information.
- It is generally faster and simpler for retrieving data.
- GET is the **default method for an HTML form** if no method is specified.

Example

```
<form action="SearchServlet">  
  <input type="text" name="query">  
  <input type="submit" value="Search">  
</form>
```

Since no `method` is specified, the browser uses **GET** by default.

3. POST Method

Definition

POST: POST is an HTTP method used to **send data to the server**, usually when submitting or processing information.

How POST Sends Data

In a POST request, the submitted data is placed inside the **HTTP request body** instead of being displayed as part of the URL.

Characteristics of POST

- Data is sent through the **request body**.
- Data is not normally displayed in the URL.
- It is suitable for sending **larger amounts of data**.
- It is commonly used when submitting forms.
- It is required for many operations involving file uploads.
- It must be explicitly specified in an HTML form.

Example:

```
<form action="LoginServlet" method="post">
  <input type="text" name="username">
  <input type="password" name="password">
  <input type="submit" value="Login">
</form>
```

Here, `method="post"` tells the browser to send the form data using POST.

4. Difference Between GET and POST

Feature	GET	POST
Purpose	Mainly used to request/retrieve data	Mainly used to submit/send data
Data Location	URL/query string	HTTP request body
URL Visibility	Data is visible in the URL	Data is not normally visible in the URL
Data Size	Suitable for relatively small data	Suitable for larger amounts of data
HTML Form Default	Default method	Must be specified using <code>method="post"</code>
Common Use	Searching, retrieving information	Login, form submission, file upload
Sensitive Data	Should not be used for passwords or other sensitive data	Preferable for sensitive form data, but HTTPS is still required

Important for Exam

POST is not automatically secure just because the data is placed in the request body.

For actual security, especially for passwords and payment information, **HTTPS** should be used.

5. GET and POST Example

Suppose a user searches for the word **Java**.

GET Request

```
https://example.com/search?query=Java
```

The search data is visible in the URL.

POST Request

```
URL:
https://example.com/search

Request Body:
query=Java
```

The data is sent inside the request body rather than being displayed in the URL.

6. Servlet Methods

In Java Servlet programming, the server needs to determine how to handle incoming HTTP requests.

The important Servlet methods are:

1. `doGet()`
 2. `doPost()`
 3. `service()`
-

7. `doGet()` Method

Definition

`doGet()` is a Servlet method used to handle an incoming **HTTP GET request**.

When a client sends a GET request to a Servlet, the Servlet container can call the Servlet's `doGet()` method.

Example

```
protected void doGet(HttpServletRequest request,
                      HttpServletResponse response)
                      throws ServletException, IOException {

    // Code for handling GET request
}
```

Working

The basic flow is:

```
Client
↓
```

```
GET Request
↓
Web Server / Servlet Container
↓
Servlet
↓
doGet()
↓
Response
↓
Client
```

Common Uses

`doGet()` can be used for:

- Retrieving information
- Searching
- Displaying data
- Requesting a resource
- Processing non-sensitive query parameters

8. `doPost()` Method

Definition

`doPost()` is a Servlet method used to handle an incoming **HTTP POST request**.

When a client sends a POST request to a Servlet, the Servlet container can call the Servlet's `doPost()` method.

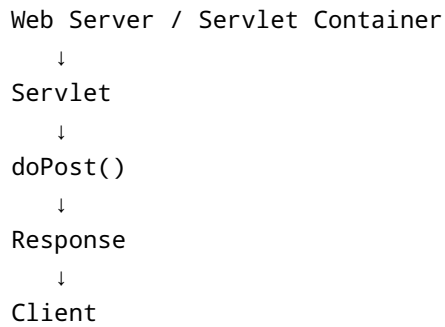
Example

```
protected void doPost(HttpServletRequest request,
                      HttpServletResponse response)
    throws ServletException, IOException {

    // Code for handling POST request
}
```

Working

```
Client
↓
POST Request
↓
```



Common Uses

`doPost()` can be used for:

- Form submission
- Login forms
- Sending larger amounts of data
- Uploading files
- Creating or processing data on the server

9. `service()` Method

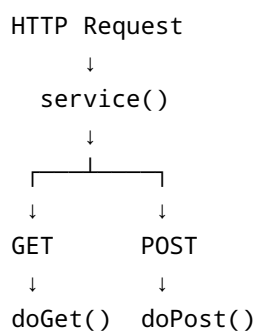
Definition

`service()` is a generic Servlet method that can handle different types of HTTP requests, including **GET** and **POST**.

In an `HttpServlet`, the `service()` method examines the HTTP request method and dispatches the request to the appropriate method such as:

- `doGet()`
- `doPost()`

Conceptually:



Important Point

Normally, developers override `doGet()` or `doPost()` rather than directly handling everything inside `service()`.

10. HTTP 405 Method Not Allowed

Definition

HTTP 405 Method Not Allowed: This error occurs when the server receives an HTTP method that the requested resource or Servlet does not allow or handle.

Example

Suppose an HTML form sends a POST request:

```
<form action="LoginServlet" method="post">
  <input type="text" name="username">
  <input type="submit">
</form>
```

But the Servlet only provides:

```
protected void doGet(HttpServletRequest request,
                      HttpServletResponse response) {

}
```

The request is **POST**, but there is no appropriate `doPost()` implementation.

This can result in:

```
HTTP 405 Method Not Allowed
```

Remember

GET request → `doGet()`

POST request → `doPost()`

The Servlet should provide the appropriate method for the HTTP request it is expected to handle.

11. GET Form vs POST Form

GET Form

```
<form action="SearchServlet" method="get">
  <input type="text" name="query">
  <input type="submit">
</form>
```

Data may appear in the URL:

SearchServlet?query=Java

POST Form

```
<form action="LoginServlet" method="post">
  <input type="text" name="username">
  <input type="password" name="password">
  <input type="submit">
</form>
```

The form data is sent in the request body.

12. When to Use GET and POST

Use GET When:

- You want to **retrieve information**.
- The data does not need to be hidden from the URL.
- The request contains relatively small amounts of data.
- You are performing operations such as searching or viewing information.

Example:

Search for "Java Servlet"

The search query can be sent as part of the URL.

Use POST When:

- You are **submitting information** to the server.
- The request contains a larger amount of data.
- You are submitting a form.
- You are uploading files.

- You are sending information that should not appear in the URL.

Example:

```
Login form
Username + Password
    ↓
  POST
    ↓
  Server
```

13. Important Exam Points

Important for Exam:

- **GET and POST are HTTP methods (HTTP verbs).**
- **GET data is normally sent through the URL/query string.**
- **POST data is sent in the HTTP request body.**
- `doGet()` **handles GET requests.**
- `doPost()` **handles POST requests.**
- `service()` **can handle both GET and POST and dispatch them appropriately.**
- If a Servlet does not support the HTTP method being requested, the server may return **HTTP 405 Method Not Allowed.**
- An HTML form uses **GET by default** when no method is specified.
- Use `method="post"` when you want the form to submit using POST.
- **POST does not by itself provide encryption or security.** Use **HTTPS** when transmitting sensitive information.

14. Commonly Confused

GET vs POST

GET: Mainly used to **get/retrieve information.**

POST: Mainly used to **send/submit information.**

`doGet()` **vs** `doPost()`

`doGet()` → Handles HTTP GET requests.

`doPost()` → Handles HTTP POST requests.

POST vs HTTPS

These are **not the same thing**.

- **POST** determines how data is transmitted in the HTTP request.
- **HTTPS** provides encryption for data transmitted between the client and server.

Therefore, sensitive information such as passwords should be sent using **POST over HTTPS**.

15. Quick Revision Table

Concept	Meaning
HTTP Method	Defines the type of action a client wants to perform
GET	Mainly used to retrieve information
POST	Mainly used to submit/send information
<code>doGet()</code>	Servlet method for GET requests
<code>doPost()</code>	Servlet method for POST requests
<code>service()</code>	Handles HTTP requests and dispatches them to methods such as <code>doGet()</code> and <code>doPost()</code>
HTTP 405	Method Not Allowed
GET Form Default	GET
POST Form	Must specify <code>method="post"</code>
HTTPS	Provides encrypted communication

16. Key Points

- **HTTP methods** allow a client and server to communicate about the action being requested.
 - **GET** is mainly used for retrieving information.
 - **POST** is mainly used for submitting or sending information.
 - GET parameters are normally visible in the **URL**.
 - POST data is normally placed in the **request body**.
 - `doGet()` handles GET requests in a Servlet.
 - `doPost()` handles POST requests in a Servlet.
 - `service()` acts as a common entry point for HTTP requests and dispatches them appropriately.
 - A mismatch between the requested HTTP method and the Servlet's supported method can produce **HTTP 405**.
 - **POST alone is not a security mechanism; HTTPS is needed for encryption.**
-

17. Summary

HTTP **GET** and **POST** are two important methods used for communication between a client and server. **GET** is mainly used to retrieve information, with parameters commonly appearing in the URL. **POST** is mainly used to submit information, with data normally placed in the request body.

In Java Servlets, `doGet()` handles GET requests and `doPost()` handles POST requests. The `service()` method can receive HTTP requests and dispatch them to the appropriate method.

For examination purposes, remember the basic relationship:

```
GET Request
  ↓
doGet()

POST Request
  ↓
doPost()

HTTP Request
  ↓
service()
  ↓
doGet() / doPost()
```

Most Important Rule:

```
GET → doGet()
POST → doPost()
```